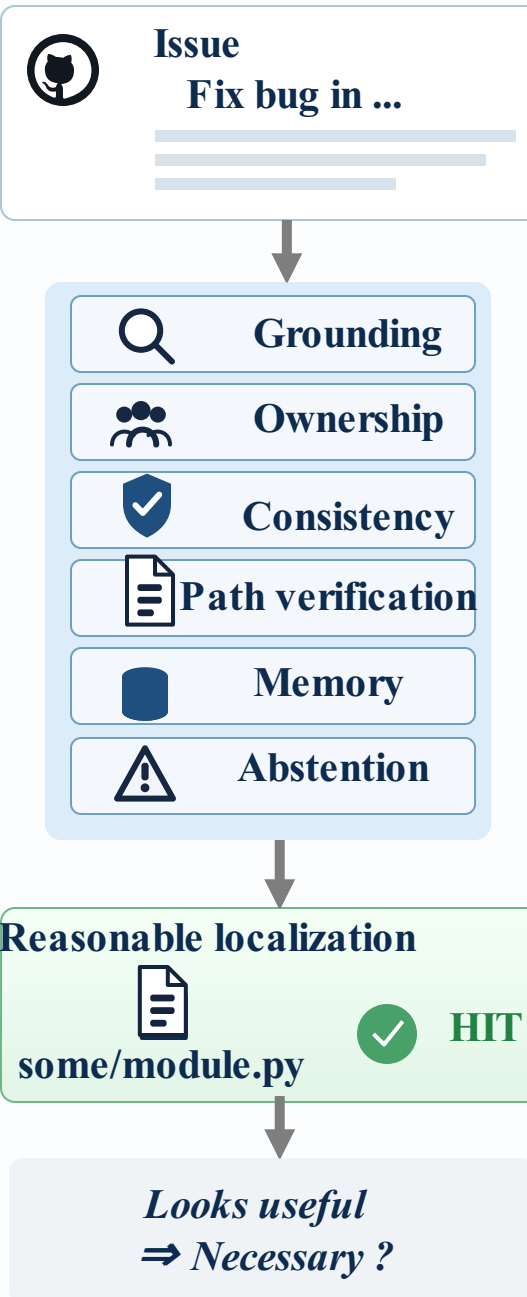


Conventional Intuition

A good-looking agent trajectory



Empirical counterexamples: matched interventions on real repository-localization cases

1. Django #14539 - context rescue

A small relation can be outcome-critical

Django #14539
Allow marking a safe string
as safe in conditional expressions ...

Structural context only

No confident candidate

django/
utils/
html.py
...

No candidate

MISS

+ Requirement /
design relation

Issue requirement
“conditional expression
marked safe”
design relation

Repository design
HTML utilities for
safe strings

django/
utils/
html.py

django/utils/html.py

HIT

46.3% → 70.4% (+24.2%)

A small, low-salience relation
can be outcome-critical.

2. Ansible password lookup

High-quality behaviors can be removable

Ansible: password lookup How
to use the password lookup
plugin to manage credentials?

Full configuration

ansible/lib/ansible/plugins/
lookup/password.py **HIT**

Single-component removals
(all retain the same top-1 file)

– Grounding	ansible/lib/.../password.py	HIT
– Ownership	ansible/lib/.../password.py	HIT
– Contradiction	ansible/lib/.../password.py	HIT
– Path verifier	ansible/lib/.../password.py	HIT
– Memory	ansible/lib/.../password.py	HIT
– Abstention	ansible/lib/.../password.py	HIT

Diagnostic value does not
imply outcome necessity.

3. System-level and adverse cases

Audit can help, but some behaviors
are adverse

(a) Ansible CPU-count
— audit rescue

No Audit
modules
/gather_facts.py
MISS

Full Audit
module_utils/facts/
hardware/linux.py
HIT

260 matched cases: 16 gains / 0 losses

(b) Ansible hashing/filter
— removal improves

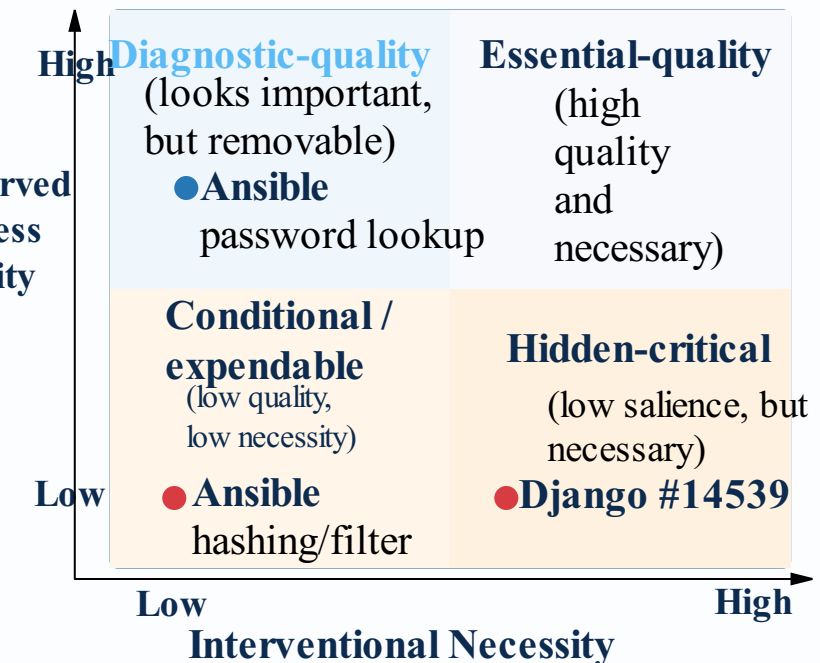
Full Audit
utils/
hashing.py
MISS

Remove selected
behaviors
plugins/
filter/core.py
HIT

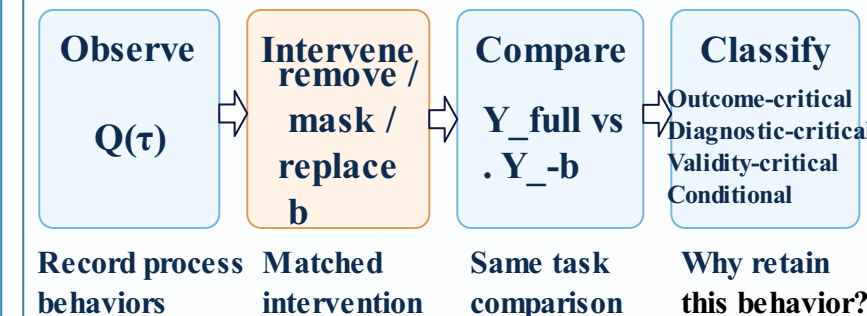
A behavior that looks desirable
can be neutral or even adverse.

Quality–Necessity Gap

Observed process quality ≠ Interventional necessity



AUDIT: measure what success depends on



AUDIT

Instrumentation + matched-intervention framework